

AI Code Review Pilot Project - Sample Data

1. Buggy C# Code Samples

Sample 1 - Null Reference Issue

```
public class Student
{
    public string Name;
    public int Marks;

    public void PrintDetails()
    {
        Console.WriteLine(Name.ToUpper());
        Console.WriteLine("Marks: " + Marks);
    }
}
```

Sample 2 - Division by Zero

```
public class Calculator
{
    public int Divide(int a, int b)
    {
        return a / b;
    }
}
```

Sample 3 - Resource Leak

```
using System.IO;

public class FileReader
{
    public string Read()
    {
        StreamReader sr = new StreamReader("data.txt");
        return sr.ReadToEnd();
    }
}
```

Sample 4 - Hardcoded Logic

```
public class Discount
{
    public double GetPrice(double price)
    {
        return price - (price * 0.10);
    }
}
```

Sample 5 - Loop Bug

```
public class Sum
{
    public int GetSum(int[] arr)
    {
        int sum = 0;
        for (int i = 1; i <= arr.Length; i++)
        {
            sum += arr[i];
        }
        return sum;
    }
}
```

2. Expected AI Output Format (JSON)

Sample 1 Output

```
{
  "issues": [
    {
      "severity": "High",
      "line": 7,
      "message": "NullReferenceException risk on Name.ToUpper()",
      "suggestion": "Check for null before using Name"
    }
  ]
}
```

Sample 2 Output

```
{
  "issues": [
    {
      "severity": "Critical",
```

```
"line": 5,  
"message": "DivideByZeroException risk",  
"suggestion": "Validate b != 0 before division"  
}  
]  
}
```

3. Edge Case Test Data

1. Empty Class:

```
public class Test {}
```

2. Multiple Bugs:

```
public class Demo  
{  
    public int Divide(int a, int b)  
    {  
        int x = a / b;  
        string s = null;  
        return s.Length + x;  
    }  
}
```

3. Clean Code:

```
public class Add  
{  
    public int Sum(int a, int b)  
    {  
        return a + b;  
    }  
}
```